# Kalai's Conjecture for Tight Trees

Dhruv Mubayi* Jacques Verstraëte†

**Abstract**

Let $r \geq 2$ and $t \geq 1$. It is shown that if $T$ is an $r$-uniform tight tree with $t$ edges and $H$ is a $T$-free $r$-uniform hypergraph, then $|E(H)| \leq (t-1)|\partial H|/r$, where $\partial H$ is the $(r-1)$-shadow of $H$. The bound is tight infinitely often, and establishes Kalai's Conjecture, whose $r = 2$ case is the Erdős-Sós Conjecture.

The proof was found by GPT-6 Astra, extending its method of proof for the Erdős-Sós conjecture to the hypergraph setting. A strengthening of the Erdős-Sós conjecture due to the authors about tight lower bounds on the number of copies of a tree in a graph with average degree $d \geq t-1 \geq 0$ remains open.

## 1 AI Declaration Statement

The proof of the main result was found by GPT-6 Astra by first prompting it to extend its recent proof of the Erdős-Sós conjecture (see Adamczewski and Bloom [6, Appendix B.4]) to the case of hypergraph tight paths as in [13], and then prompting it to prove the full Kalai conjecture. As we elaborate throughout the text, GPT-6 Astra's proof technique for Erdős-Sós conjecture, which uses permutations of the underlying vertex set, perhaps has its genesis in an old argument of Perles (see [18]), which considered convex geometric graphs (these may be viewed as graphs equipped with a cyclic ordering of their vertices). This method was developed in the hypergraph case by Füredi-Jiang-Kostochka-Mubayi-Verstraëte in [13] and gave the previously best bounds for Kalai's conjecture for tight paths. The authors have checked the proof and rewritten much of the text provided by GPT-6 Astra and added further explanations (and pictures) throughout the paper. They take full responsibility for all the content in the paper.

## 2 Introduction

All hypergraphs in this paper are finite and simple. An $r$-uniform hypergraph, or *$r$-graph*, has edges of size $r$. For an $r$-graph $T$, let $\mathrm{ex}(n, T)$ denote the maximum number of edges in an $n$-vertex $r$-graph containing no copy of $T$.

The Erdős–Gallai Theorem [8] asserts that a graph with no path of $t$ edges has at most $(t-1)n/2$ edges. The Erdős–Sós Conjecture [7] asks for the same bound when the forbidden path is replaced by

*Department of Mathematics, Statistics and Computer Science, University of Illinois, Chicago, IL 60607. Email: mubayi@uic.edu. Research partially supported by NSF Awards DMS-2552740 and DMS-2153576.

†Department of Mathematics, University of California, San Diego, CA, 92093-0112 USA. Email: jverstraete@ucsd.edu. Research supported by NSF-BSF award DMS-2347832.

an arbitrary tree with $t$ edges. This conjecture has received extensive attention in the literature [1–5, 7–10, 15, 16, 19, 21, 22, 24–26, 28, 30]. The Erdős–Sós Conjecture was recently proved employing an argument using permutations of the vertex set by GPT-6 Astra AI, described by Adamczewski and Bloom [6, Appendix B.4]. Similar counting arguments are key to the method of Perles [18] giving a very elegant alternative proof of the Erdős–Gallai Theorem using convex geometric graphs, and to extensions to uniform hypergraphs in [13].

Kalai proposed a hypergraph generalization of the Erdős–Sós Conjecture in 1984, recorded by Frankl and Füredi [11], using the following notion of a tree. An $r$-graph $T$ is a *tight $r$-tree* if its edges have an ordering $e_1, \dots, e_t$ such that for every $i > 1$, there exists $z_i \in e_i$ and an index $p(i) < i$ satisfying

$$z_i \notin \bigcup_{h<i} e_h, \qquad e_i \setminus \{z_i\} \subseteq e_{p(i)}. \tag{2.1}$$

We take $V(T) = \bigcup_{e\in E(T)} e$, so a tight $r$-tree with $t$ edges has exactly $t + r - 1$ vertices – the case $r = 2$ is precisely a tree in the graph sense. Kalai's Conjecture is as follows:

**Conjecture 1. (Kalai's Conjecture)** *For $n \geq r \geq 2$ and every tight $r$-tree $T$ with $t$ edges,*

$$\mathrm{ex}(n, T) \leq \frac{t-1}{r}\binom{n}{r-1}. \tag{2.2}$$

To motivate this conjecture, let $b = t + r - 2$ and take a family of $b$-vertex blocks in which each $(r-1)$-set lies in at most one block and almost every $(r-1)$-set is covered, and put a complete $r$-graph on each block. Any copy of a tight tree must stay in a single block, since two edges from different blocks cannot meet in $r-1$ vertices. It therefore has at most $b = t+r-2$ vertices and hence at most $t-1$ edges. Using Rödl's packing theorem [23] providing the existence of asymptotically optimal partial Steiner systems, one obtains as $n \to \infty$:

$$\mathrm{ex}(n, T) \geq \left(\frac{t-1}{r} - o(1)\right) \cdot \binom{n}{r-1}.$$

Keevash's existence theorem for designs [17] (see also Glock, Kühn, Lo and Osthus [14]) gives exact equality in (2.2) for infinitely many $n$ by covering every $(r-1)$-set exactly once with $b$-sets – these are Steiner systems $S(r-1, b, n)$. In the special case that the tree $T$ is a sunflower – every edge of $T$ has the form $R \cup \{v\}$ where $|R| = r - 1$ – then any $r$-uniform hypergraph in which every $(r-1)$-set is contained in exactly $t-1$ hyperedges is $T$-free. These are sometimes called *simple $(r-1)$-$(n, r, t-1)$-designs*.

Several cases of Kalai's Conjecture were previously known. Frankl and Füredi [11] proved (2.2) for star-shaped tight trees, which have an edge meeting every other edge in $r-1$ vertices. Füredi, Jiang, Kostochka and the authors [12] proved an asymptotic version for tight trees with a bounded trunk and the exact bound for tight trees with at most four edges. Here a trunk is an initial tight subtree in a construction (2.1) such that each remaining edge attaches directly to an edge of that subtree; the asymptotic statement fixes $r$ and the trunk size and lets $t$ tend to infinity. Stein [27] proved (2.2) for every tight $r$-tree when the host $r$-graph is $r$-partite.

## 2.1 Main Theorem

The *shadow* of an $r$-graph $H$ is

$$\partial H = \{e \subseteq V(H) : |e| = r - 1,\ e \subseteq h \text{ for some } h \in E(H)\}.$$

We prove Kalai's conjecture by proving the following shadow bound (the equivalence to Kalai's Conjecture was established by Füredi, Jiang, Kostochka and the authors [12, Proposition 2.2]).

**Theorem 2.1.** *Let $n \geq r \geq 2$, and let $T$ be a tight $r$-tree with $t \geq 1$ edges. Every $n$-vertex $T$-free $r$-graph $H$ satisfies*

$$|E(H)| \;\leq\; \frac{t-1}{r} \cdot |\partial H| \;\leq\; \frac{t-1}{r} \cdot \binom{n}{r-1}. \tag{2.3}$$

This theorem also settles Conjecture 1.1 of Füredi, Jiang, Kostochka and the authors [13]. The proof is given in Section 4.

GPT-6 Astra has supplied a proof of the following characterization of equality. Suppose $t \geq 2$, put $b = t + r - 2$, and let $H$ be a $T$-free $r$-graph. If $T$ is not a sunflower, then equality in the shadow bound $r|E(H)| = (t-1)|\partial H|$ holds if and only if $E(H) = \bigcup_{B \in \mathcal{B}} \binom{B}{r}$, where $\mathcal{B}$ is a family of $b$-vertex blocks such that distinct blocks intersect in at most $r-2$ vertices. Equality in the binomial bound

$$|E(H)| = \frac{t-1}{r}\binom{n}{r-1}$$

holds if and only if these blocks form a Steiner system $S(r-1, b, n)$. If $T$ is a sunflower, equality in the shadow bound holds if and only if every member of $\partial H$ is contained in exactly $t-1$ edges of $H$; equality in the binomial bound requires this condition for every $(r-1)$-subset of $V(H)$. We have not verified the proof by Astra of this characterization.

## 2.2 Counting conjecture

The authors [20] proposed the following counting conjecture: for every integer $t \geq 1$, there exists $d_0 = d_0(t)$ such that every $n$-vertex graph $G$ of average degree $d \geq d_0$ satisfies

$$N_T(G) \geq n(d)_t \tag{2.4}$$

for every tree $T$ with $t$ edges. Here $N_T(G)$ denotes the number of injective graph homomorphisms from $T$ to $G$, and

$$(x)_t = x(x-1)\cdots(x-t+1)$$

for real $x$. For $d > t$, the authors [20] proved the asymptotically sharp estimate

$$N_T(G) \geq \left(1 - O\left(\frac{t^5}{d^2}\right)\right) n(d)_t. \tag{2.5}$$

Wilson [29] subsequently proved the conjectured bound (2.4) with $d_0(t) = O(t^4)$. A natural strengthening posed by the authors is that one can take

$$d_0(t) = t - 1.$$

This would imply the Erdős–Sós Conjecture, since $(d)_t > 0$ whenever $d > t-1$. The strengthened bound can be verified for certain small trees and for double stars. GPT-6 Astra has claimed proofs of (2.4) for $d \geq Ct^3$ for arbitrary trees and for $d \geq Ct^2$ for paths, where $C > 0$ is an absolute constant. We have not verified these proofs.

We propose the following hypergraph generalization. For $r$-graphs $T$ and $H$, let $N_T(H)$ denote the number of injective maps $\varphi : V(T) \to V(H)$ such that $\varphi(e) \in E(H)$ for every $e \in E(T)$.

**Conjecture 2.** *Let $n, r, t$ be integers with $n \geq r \geq 2$ and $t \geq 1$, and let $d \geq t-1$ be real. If $H$ is an $n$-vertex $r$-graph with at least*

$$\frac{d}{r}\binom{n}{r-1}$$

*edges, then, for every tight $r$-tree $T$ with $t$ edges,*

$$N_T(H) \geq (r-1)!\binom{n}{r-1}(d)_t. \tag{2.6}$$

For an $(r-1)$-set $A \subseteq V(H)$, let $d_H(A)$ denote the number of edges of $H$ containing $A$. The average $(r-1)$-degree of $H$ is

$$\frac{1}{\binom{n}{r-1}} \sum_{A \in \binom{V(H)}{r-1}} d_H(A) = \frac{r|E(H)|}{\binom{n}{r-1}},$$

where the average includes sets of codegree zero. Thus Conjecture 2 asserts that an $r$-graph of average $(r-1)$-degree $d \geq t-1$ contains at least $(r-1)!\binom{n}{r-1}(d)_t$ labelled copies of every prescribed tight $r$-tree with $t$ edges. It would imply Kalai's Conjecture (Conjecture 1), again because $(d)_t > 0$ when $d > t-1$, but remains open even for paths in graphs. GPT-6 Astra has also claimed a proof of Conjecture 2 for $d > 16r^2t^3$. We have not verified this proof.

The bound in Conjecture 2 is sharp whenever there exists a Steiner system $S(r-1, d+r-1, n)$, with $d \geq t$ an integer. Put $b = d+r-1$, let $\mathcal{B}$ be its block family, and define $E(H) = \bigcup_{B \in \mathcal{B}} \binom{B}{r}$. Every $(r-1)$-set has codegree $d$, so $|E(H)| = \frac{d}{r}\binom{n}{r-1}$. Moreover, every copy of a tight tree lies in a single block, since edges belonging to distinct blocks cannot meet in $r-1$ vertices. As a tight $r$-tree with $t$ edges has $t+r-1$ vertices, we obtain

$$N_T(H) = \frac{\binom{n}{r-1}}{\binom{b}{r-1}}(b)_{t+r-1} = (r-1)!\binom{n}{r-1}(d)_t.$$

### 2.3 Organization

The special case of trees that are *tight paths* contains the main ideas of the proof of Theorem 2.1, and is presented in Section 3. This proof uses the broad framework that was employed in [13], which previously had the best bound for the extremal number of tight paths. The proof of Theorem 2.1 is given in subsequent sections.

## 3 Proof of Theorem 2.1 for tight paths

The *tight path* $P_k^r$ has vertex set $\{x_1, \ldots, x_{k+r-1}\}$ and edges $\{x_i, \ldots, x_{i+r-1}\}$ for $1 \leq i \leq k$. Using convex geometric hypergraphs, Füredi, Jiang, Kostochka and the authors [13, Theorem 1.2] proved

$$\mathrm{ex}(n, P_k^r) \leq \begin{cases} \dfrac{k-1}{2}\binom{n}{r-1}, & r \text{ even}, \\ \dfrac{1}{2}\left(k + \left\lfloor \dfrac{k-1}{r} \right\rfloor\right)\binom{n}{r-1}, & r \text{ odd}. \end{cases} \tag{3.1}$$

The proof of this theorem, which proceeds by considering a count of ordered tight paths in so-called convex geometric hypergraphs [13], partly motivates the proof of Theorem 2.1; indeed in this section we prove the tight path case of Theorem 2.1, as it is much simpler and yet illustrates the main ideas. Roughly speaking, we count permutations whose first $r-1$ vertices, together with a marked later vertex, form an edge of the host hypergraph. For paths, a cyclic rotation extends every eligible state except the first one in each permutation, and these ideas were used to study the extremal function for tight paths in convex geometric hypergraphs [13]; we give this simpler "path argument" proof of Theorem 2.1 in this section. The more complicated case of general trees is achieved in Section 4.

### 3.1 Definitions and notation

Fix an $r$-graph $H$ on a vertex set $V$ of size $n \geq r$. For a permutation $\pi = (v_1, \ldots, v_n)$ of $V$, and $j \in [n]$, write

$$\mathrm{Init}(\pi) = (v_1, \ldots, v_{r-1}), \qquad \pi_{\leq j} = \{v_1, \ldots, v_j\}.$$

Let

$$\mathcal{W} = \{\pi : \pi_{\leq r-1} \in \partial H\}$$

be the set of permutations whose first $r-1$ members are in the shadow of $H$, and define the set of *marked states* by

$$\Omega = \big\{(\pi, j) : \pi \in \mathcal{W},\ r \leq j \leq n,\ \{v_1, \ldots, v_{r-1}, v_j\} \in E(H)\big\}; \tag{3.2}$$

we say that $v_j$ is the marked vertex of the edge $\{v_1, \ldots, v_{r-1}, v_j\}$. Set $W = |\mathcal{W}|$ and $M = |\Omega|$. Direct counting gives

$$\begin{aligned} W &= (r-1)!\,|\partial H|\,(n-r+1)!, \\ M &= r!\,|E(H)|\,(n-r+1)!. \end{aligned} \tag{3.3}$$

For the first identity, choose the initial shadow edge, order it, and order the remaining vertices. For the second, choose an edge, choose its marked vertex in $r$ ways, and order its other $r-1$ vertices at the beginning of the permutation. Order all remaining $n-r+1$ vertices arbitrarily; the position of the chosen marked vertex determines $j$. For $\ell \geq 1$, let $\mathcal{S}_\ell \subseteq \Omega$ consist of the states $(\pi, j)$ for which $H[\pi_{\leq j}]$ contains a tight path of $\ell$ edges ending with the ordered tuple $\mathrm{Init}(\pi)$. Explicitly, there must be distinct vertices $x_1, \ldots, x_{\ell+r-1} \in \pi_{\leq j}$ such that

$$\text{For } 1 \leq i \leq \ell, \{x_i, \ldots, x_{i+r-1}\} \in E(H) \text{ and } (x_{\ell+1}, \ldots, x_{\ell+r-1}) = \mathrm{Init}(\pi). \tag{3.4}$$

The permutation specifies the allowed vertex set and the terminal tuple; the path may traverse its other vertices in any order. Every marked state supports the one-edge path $(v_j, v_1, \ldots, v_{r-1})$, so

$$\mathcal{S}_1 = \Omega. \tag{3.5}$$

### 3.2 Proof for tight paths

**Lemma 3.1.** *For every $\ell \geq 1$,*

$$|\mathcal{S}_\ell| \leq |\mathcal{S}_{\ell+1}| + W. \tag{3.6}$$

*Proof.* For each $\pi \in \mathcal{W}$ having a state in $\mathcal{S}_\ell$, discard $(\pi, j)$ with $j$ least among its states in $\mathcal{S}_\ell$. If $j \neq j'$, we do not discard both $(\pi, j)$ and $(\pi, j')$ as $j < j'$ or $j' < j$. Hence we can map a discarded state $(\pi, j)$ to $\pi \in \mathcal{W}$ and this map is injective. Consequently, at most $|\mathcal{W}| = W$ states are discarded. We provide an injection $\Phi'$ from the remaining states into $\mathcal{S}_{\ell+1}$.

For any $(\pi, j) \in \Omega$, write

$$\pi = (a_1, \dots, a_{r-1}, A, x, B), \qquad x = v_j,$$

where $A = (v_r, \dots, v_{j-1})$ and $B = (v_{j+1}, \dots, v_n)$ are possibly empty blocks. Define

$$\Phi(\pi, j) = (\pi', j), \qquad \pi' = (a_2, \dots, a_{r-1}, x, A, a_1, B). \tag{3.7}$$

This rotates the entries in positions $1, \dots, r-1, j$ and fixes all other entries. It preserves the marked edge, and hence maps $\Omega$ to itself. Since $\Phi^r$ is the identity, $\Phi$ is a bijection of $\Omega$. Moreover,

$$\pi'_{\le j} = \pi_{\le j}, \qquad \mathrm{Init}(\pi') = (a_2, \dots, a_{r-1}, x). \tag{3.8}$$

Now let $(\pi, j) \in \mathcal{S}_\ell$ be a state that has not been discarded in the first step. There exists $i < j$ with $(\pi, i) \in \mathcal{S}_\ell$. A witnessing path for this earlier state lies in $\pi_{\le i}$ and ends with $(a_1, \dots, a_{r-1})$. Since $x = v_j \notin \pi_{\le i}$, appending $x$ introduces a new vertex and the new edge $\{a_1, \dots, a_{r-1}, x\} \in E(H)$. The resulting tight path has $\ell + 1$ edges, lies in $\pi'_{\le j}$, and ends with $\mathrm{Init}(\pi')$. Thus $\Phi(\pi, j) \in \mathcal{S}_{\ell+1}$. The restriction of the bijection $\Phi$ is the required injection $\Phi'$. □

*Proof of Theorem 2.1 for tight paths.* We aim to show that if $H$ is an $n$-vertex $r$-uniform $P_k^r$-free hypergraph, then $|E(H)| \le (k-1)/r \cdot |\partial H|$. The assertion is immediate if $E(H) = \varnothing$. Otherwise $n \ge r$, so the preceding definitions apply. If $H$ is $P_k^r$-free, then $\mathcal{S}_k = \varnothing$. Iterating Lemma 3.1 and using (3.5) gives

$$M = |\mathcal{S}_1| \le (k-1)W.$$

Substituting (3.3) and cancelling the factor $(r-1)!(n-r+1)!$ yields $r|E(H)| \le (k-1)|\partial H|$. □

# 4 Proof of Theorem 2.1

For general tight trees, we adapt the permutation framework of Adamczewski and Bloom [6, Appendix B.4] by prescribing the images of all vertices of a root edge. Our inductive estimate is proved for every choice of root edge: after deleting a leaf edge, we apply induction with the attaching edge of the remaining tree as its root, and this edge need not itself be a leaf in the smaller tree (if the tree is a path it will be a leaf). To handle a non-leaf root, we apply induction using two smaller rooted trees meeting exactly in their common root edge. The path-extension argument in Section 3 does not require this joining step, since paths can be built by repeatedly adding an edge at an end.

We retain the notation $H$, $V$, $\mathcal{W}$, $\Omega$, $W$ and $M$ from Section 3. The proof for trees uses embeddings with a prescribed image of a whole root edge. The following straightforward decomposition lemma allows us to do induction on the size of the tree. Its proof is trivial when $r = 2$ and only slightly more complicated when $r > 2$.

**Lemma 4.1.** *Let $r \ge 2$ and $T$ be a tight $r$-tree with at least two edges, and fix $e \in E(T)$. At least one of the following holds.*

(i) *$T$ is obtained from a smaller tight tree $S$ by adding a new vertex $z$ and the edge $e = F \cup \{z\}$, where $F$ is an $(r-1)$-subset of an edge $f$ of $S$.*

(ii) *$T = T_1 \cup T_2$ for tight trees $T_1, T_2$, each with fewer edges than $T$, such that*

$$E(T_1) \cap E(T_2) = \{e\}, \qquad V(T_1) \cap V(T_2) = e.$$

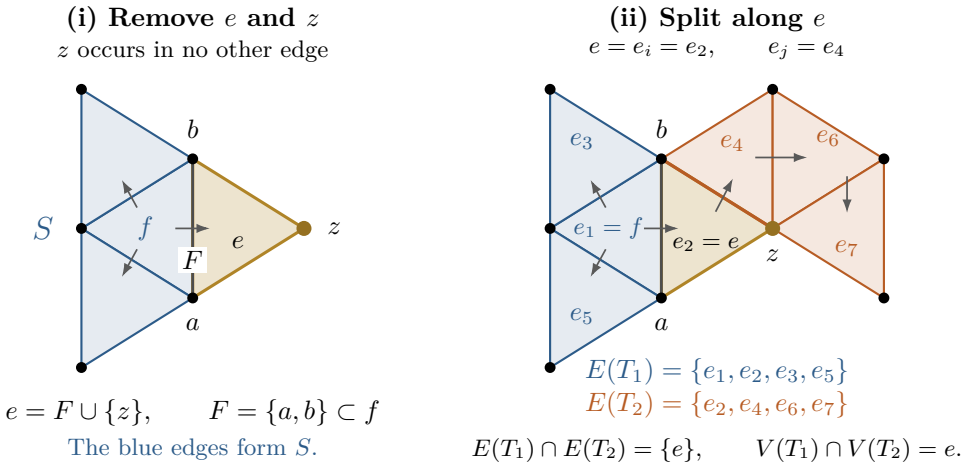


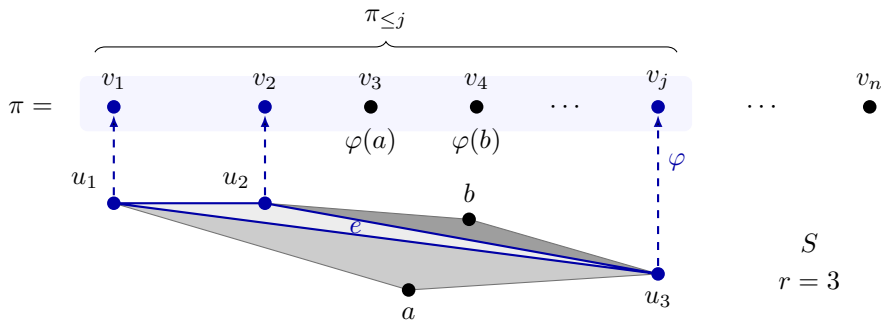

Figure 1: The two cases of Lemma 4.1, illustrated for $r = 3$.

Here is a sketch of a proof of Lemma 4.1. Choose a construction order $e_1, \dots, e_t$ of $T$ with $e = e_i$ and $i > 1$, interchanging the first two edges if $e = e_1$. Let $z$ be the unique vertex of $e$ outside $e_1 \cup \cdots \cup e_{i-1}$. If no other edge contains $z$, deleting $e$ and $z$ gives the tree $S$ in (i). Otherwise, let $j > i$ be the least index such that $z \in e_j$. Initially, let $T_1$ consist of $e_1, \dots, e_{j-1}$ and let $T_2$ consist of $e, e_j$. Process the remaining edges in the construction order. For each $e_k$, choose an earlier edge containing its $r-1$ previously introduced vertices, and add $e_k$ to the tree containing that earlier edge. If the chosen edge is $e$, add $e_k$ to either tree, but not both. The resulting trees give (ii). See Figure 1 for an example.

## 4.1 States rooted at an edge

We now adapt the definition of $\mathcal{S}_\ell$ in Section 3.1 from tight paths to arbitrary hypergraphs, prescribing the images of all vertices of a root edge. Let $S$ be an $r$-graph with a distinguished ordered edge $\vec{e} = (u_1, \dots, u_r)$. For each permutation $\pi = (v_1, \dots, v_n)$ of $V(H)$, a state $(\pi, j) \in \Omega$ is *good for* $(S, \vec{e})$ if there is an injective embedding $\varphi : V(S) \longrightarrow V(H)$ such that for $1 \le i \le r-1$,

$$\begin{gathered} \varphi(u_i) = v_i, \qquad \varphi(u_r) = v_j, \\ \varphi(V(S)) \subseteq \pi_{\le j}. \end{gathered} \tag{4.1}$$




A good state for $(S, (u_1, u_2, u_3))$. Shaded triangles represent hyperedges, and $j > 4$. The root maps to $(v_1, v_2, v_j)$, while the entire image of $S$ lies in $\pi_{\le j}$.

Here an embedding sends every edge of $S$ to an edge of $H$. Let $C(S, \vec{e})$ be the number of good states. Each state is counted once, regardless of how many embeddings witness it.

Observe that $C(S, \vec{e})$ does not depend on the ordering of $e$. Indeed, write $q_i = i$ for $i < r$ and $q_r = j$. If the root is reordered as $(u_{\sigma(1)}, \dots, u_{\sigma(r)})$, permute the host entries in these positions so that

$$v'_{q_i} = v_{q_{\sigma(i)}} \qquad (1 \le i \le r),$$

and keep all other entries fixed. This is a bijection on $\Omega$, preserves $\pi_{\le j}$, and makes the same embedding a witness for the reordered root. We therefore write $C(S, e)$, choosing an ordering of the fixed edge $e$ when convenient. The count can depend on the choice of $e$. For the one-edge tree,

$$C(e, e) = M. \tag{4.2}$$

## 4.2 Adding a leaf

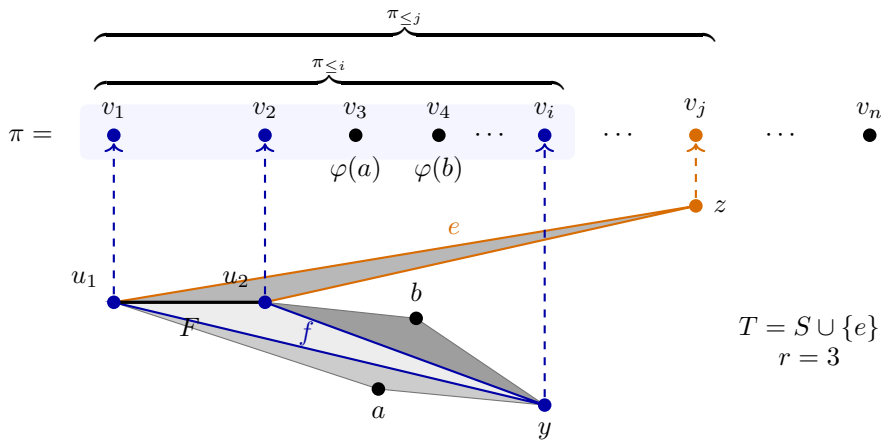
The following lemma deals with Case (i) of Lemma 4.1 in the decomposition of a tight tree, and is the analog of the case of paths (Lemma 3.1) in Section 3.

**Lemma 4.2.** *Let $S$ be an $r$-graph, let $f \in E(S)$, and let $F \subset f$ have size $r - 1$. Form $T$ by adding a new vertex $z$ and the edge $e = F \cup \{z\}$. Then*

$$C(S, f) \le C(T, e) + W. \tag{4.3}$$

*Proof.* Write $F = \{u_1, \dots, u_{r-1}\}$ and $f = F \cup \{y\}$. Choose the root orders $(u_1, \dots, u_{r-1}, y)$ for $S$ and $(u_1, \dots, u_{r-1}, z)$ for $T$. For every $\pi \in \mathcal{W}$, discard its first good state for $(S, f)$, if one exists. This discards at most $W$ states.

For a retained state $(\pi, j)$, there is an earlier good state $(\pi, i)$ with $i < j$. Choose its witnessing embedding $\varphi$ of $S$. It maps $u_h$ to $v_h$ for $h < r$, maps $y$ to $v_i$, and has image in $\pi_{\le i}$. Since $v_j \notin \pi_{\le i}$, extend $\varphi$ by sending $z$ to $v_j$. The additional edge maps to $\{v_1, \dots, v_{r-1}, v_j\} \in E(H)$, and the resulting embedding satisfies (4.1) for $(T, e)$ at $(\pi, j)$. Thus every retained state is itself good for $(T, e)$, proving (4.3). □




Adding a leaf: $f = \{u_1, u_2, y\}$ is the old root, and $e = \{u_1, u_2, z\}$ is the new edge. The earlier embedding of $S$ lies in $\pi_{\le i}$, with $\varphi(y) = v_i$; it extends by setting $\varphi(z) = v_j$, where $i < j$.

### 4.3 Joining along root edges

The more complicated decomposition offered by Case (ii) of Lemma 4.1 is the subject of this section. For a finite set $U$, let $\mathrm{Perm}(U)$ denote the set of permutations of $U$. For $w = (w_1, \dots, w_m) \in \mathrm{Perm}(U)$, write

$$w_{\le s} = (w_1, \dots, w_s), \qquad w_{>s} = (w_{s+1}, \dots, w_m).$$

We note that we have changed the definition of $w_{\le s}$ from previous sections to denote an initial part of the permutation rather than the set of elements in this initial part. The support of a permutation or one of its blocks is the set of its entries. When $|U| = m$ and $\mathcal{F} \subseteq 2^U$, put

$$C_U(\mathcal{F}) = \{(w, s) : w \in \mathrm{Perm}(U),\ 0 \le s \le m,\ \mathrm{supp}(w_{\le s}) \in \mathcal{F}\}$$

and $c_U(\mathcal{F}) = |C_U(\mathcal{F})|$. The value $s = 0$ is allowed and corresponds to the empty prefix. Thus $(w, 0) \in C_U(\mathcal{F})$ if and only if $\varnothing \in \mathcal{F}$. The ambient set of all pairs $(w, s)$, where $w \in \mathrm{Perm}(U)$ and $0 \le s \le m$, has $(m+1)m! = (m+1)!$ elements.

To motivate Lemma 4.3, consider its application to a tight tree $T = T_1 \cup T_2$, where $V(T_1) \cap V(T_2) = e$ and $e$ is an edge of both subtrees. Fix an assignment of the vertices of $e$ to an edge of $H$, let $A_0$ be its image, and put $U = V(H) \setminus A_0$. The families $\mathcal{A}$, $\mathcal{B}$, and $\mathcal{C}$ consist of those sets $X \subseteq U$ for which $T_1$, $T_2$, and $T$, respectively, embed into $H[A_0 \cup X]$ with the prescribed assignment on $e$. Thus $X$ specifies the vertices available outside the root edge.

**Lemma 4.3.** *Suppose $\mathcal{A}, \mathcal{B}, \mathcal{C} \subseteq 2^U$ satisfy $\mathcal{C} \subseteq \mathcal{A}$ and*

$$R \in \mathcal{A}, \quad X \in \mathcal{B}, \quad R \cap X = \varnothing \quad \Longrightarrow \quad R \cup X \in \mathcal{C}. \tag{4.4}$$

*If $|U| = m$, then*

$$c_U(\mathcal{A}) + c_U(\mathcal{B}) \le (m+1)! + c_U(\mathcal{C}). \tag{4.5}$$

*Proof.* We construct an injection from the set of pairs $(w, s) \in C_U(\mathcal{A})$ for which $\mathrm{supp}(w_{\le s}) \notin \mathcal{C}$ to the set of all pairs $(w', s')$ with $\mathrm{supp}(w'_{\le s'}) \notin \mathcal{B}$. This will imply $c_U(\mathcal{A}) - c_U(\mathcal{C}) \le (m+1)! - c_U(\mathcal{B})$ as required.

Let $(w, s)$ be a pair of the first kind, and let $R$ be the shortest prefix of $w$ with $\mathrm{supp}(R) \in \mathcal{A} \setminus \mathcal{C}$. This prefix exists and has length at most $s$. Define $X$ and $Y$ by writing

$$w = RXY, \qquad s = |R| + |X|, \qquad w' = XRY, \qquad s' = |X|$$

where juxtaposition denotes concatenation, and define

$$(w,\, s) \longmapsto (w',\, s'). \tag{4.6}$$

Figure 2 illustrates this map. We move the initial block $R$ past $X$, preserving the order within each block. The marked prefix changes from $RX$ in the top row to $X$ in the bottom row, so the marked position changes from $s = |R| + |X|$ to $s' = |X|$.

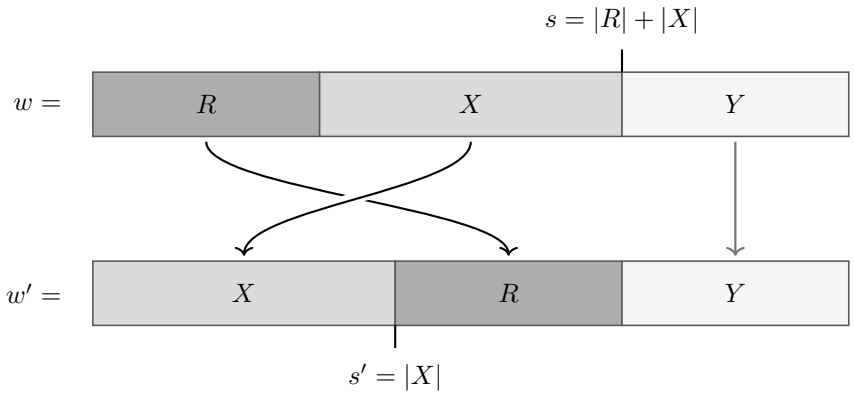



Figure 2: The map $(w, s) \mapsto (w', s')$. The blocks $R$ and $X$ exchange places, while $Y$ stays fixed.

If $\mathrm{supp}(X) \in \mathcal{B}$, then (4.4) would imply $\mathrm{supp}(RX) \in \mathcal{C}$, contrary to the assumption on the input pair. Hence the output has prefix outside $\mathcal{B}$.

Given an output pair $(w', s')$, we now show how to recover $(w, s)$ uniquely. First observe that, given $w'$, the integer $s'$ determines the word $X$. In the remaining suffix $w'_{>s'} = RY$, take the shortest prefix whose support belongs to $\mathcal{A} \setminus \mathcal{C}$. This prefix is exactly $R$: we have $\mathrm{supp}(R) \in \mathcal{A} \setminus \mathcal{C}$, and every shorter prefix of $R$ has support outside $\mathcal{A} \setminus \mathcal{C}$ by the minimality of $R$ as a prefix of the original word $w$. We thereby recover $R$ and $Y$, and hence $(w, s)$. This also applies when $R$ or $X$ is empty. □

In the application to tight trees below, we have the stronger inclusion $\mathcal{C} \subseteq \mathcal{A} \cap \mathcal{B}$: an embedding of the whole tree restricts to an embedding of either rooted subtree. However, the lemma requires only $\mathcal{C} \subseteq \mathcal{A}$.

**Lemma 4.4.** *Suppose $T = T_1 \cup T_2$, where*

$$V(T_1) \cap V(T_2) = e, \qquad e \in E(T_1) \cap E(T_2).$$

*Then*

$$C(T_1, e) + C(T_2, e) \le M + C(T, e). \tag{4.7}$$

*Proof.* We prove the inequality separately for each prescribed image of the ordered root edge, and then sum over these assignments. Fix an ordering $e = (u_1, \dots, u_r)$, and let $\mathcal{I}_H$ be the set of ordered edges of $H$, so that $|\mathcal{I}_H| = r!|E(H)|$. For each $\mathbf{a} = (a_1, \dots, a_r) \in \mathcal{I}_H$, put

$$\Omega_{\mathbf{a}} = \{(\pi, j) \in \Omega : v_i = a_i \text{ for } 1 \le i < r, \quad v_j = a_r\},$$

where $\pi = (v_1, \ldots, v_n)$. Let $C_{\mathbf{a}}(S, e)$ be the number of states in $\Omega_{\mathbf{a}}$ that are good for $(S, e)$. Thus $(\pi, j) \in \Omega_{\mathbf{a}}$ precisely when $S$ admits an embedding into $H[\{v_1, \ldots, v_j\}]$ that maps $u_i$ to $a_i$ for every $1 \leq i \leq r$.

Every state $(\pi, j) \in \Omega$ determines the unique assignment $\mathbf{a} = (v_1, \ldots, v_{r-1}, v_j)$. Thus the sets $\Omega_{\mathbf{a}}$ partition $\Omega$, and

$$C(S, e) = \sum_{\mathbf{a} \in \mathcal{I}_H} C_{\mathbf{a}}(S, e) \qquad \text{for } S \in \{T_1, T_2, T\}. \tag{4.8}$$

We will show that, for every $\mathbf{a} \in \mathcal{I}_H$,

$$C_{\mathbf{a}}(T_1, e) + C_{\mathbf{a}}(T_2, e) \leq |\Omega_{\mathbf{a}}| + C_{\mathbf{a}}(T, e). \tag{4.9}$$

Summing these inequalities will give the desired result.

Fix $\mathbf{a} = (a_1, \ldots, a_r) \in \mathcal{I}_H$, and put

$$A_0 = \{a_1, \ldots, a_r\}, \qquad U = V(H) \setminus A_0, \qquad m = n - r.$$

Let $\psi : e \longrightarrow A_0$ be the bijection defined by $\psi(u_i) = a_i$ for $1 \leq i \leq r$. Define $\mathcal{A}$, $\mathcal{B}$ and $\mathcal{C}$ to be the families of sets $Y \subseteq U$ for which $T_1$, $T_2$ and $T$, respectively, admit an embedding into $H[A_0 \cup Y]$ whose restriction to $e$ is $\psi$. We first check that these families satisfy the hypotheses of Lemma 4.3. If $Y \in \mathcal{C}$, restrict a witnessing embedding of $T$ to $V(T_1)$. This gives an embedding of $T_1$ into $H[A_0 \cup Y]$ with the same assignment on $e$. Hence $Y \in \mathcal{A}$, proving $\mathcal{C} \subseteq \mathcal{A}$. Now suppose that $R \in \mathcal{A}$, $X \in \mathcal{B}$ and $R \cap X = \varnothing$. Choose witnessing embeddings $\varphi_1$ and $\varphi_2$. They agree on $e$, where both equal $\psi$, and

$$\varphi_1(V(T_1) \setminus e) \subseteq R, \qquad \varphi_2(V(T_2) \setminus e) \subseteq X.$$

Their images outside $A_0$ are therefore disjoint. Since $V(T_1) \cap V(T_2) = e$, their union is an injective embedding of $T$ into $H[A_0 \cup R \cup X]$. Thus $R \cup X \in \mathcal{C}$, as illustrated in Figure 3. This verifies the hypotheses of Lemma 4.3.

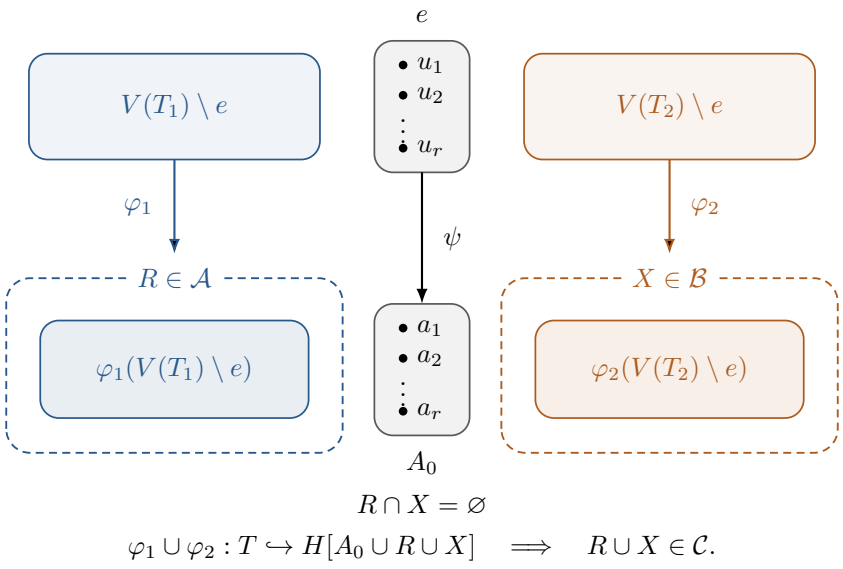



Figure 3:

We will show that $C_{\mathbf{a}}(T_1, e) = c_U(\mathcal{A}), C_{\mathbf{a}}(T_2, e) = c_U(\mathcal{B})$, and $C_{\mathbf{a}}(T, e) = c_U(\mathcal{C})$. We prove the first equality by constructing a bijection between the good states counted by $C_{\mathbf{a}}(T_1, e)$ and the pairs

counted by $c_U(\mathcal{A})$. The other two equalities follow by the same argument. Recall that $c_U(\mathcal{A})$ counts pairs $(w, s)$ with $w \in \mathrm{Perm}(U)$ and $0 \le s \le m$ such that $\{w_1, \dots, w_s\} \in \mathcal{A}$. Take a state $(\pi, j) \in \Omega_{\mathbf{a}}$ that is good for $(T_1, e)$, and write $\pi = (v_1, \dots, v_n)$. The vertices $a_1, \dots, a_{r-1}$ occupy the first $r-1$ positions of $\pi$, and $a_r$ occupies the marked position $j$. Delete these $r$ vertices, preserving the order of all remaining vertices, to obtain a permutation $w$ of $U$. Put $s = j - r$. Exactly $s$ of the first $j$ vertices of $\pi$ lie outside $A_0$, and these become the first $s$ entries of $w$. Consequently,

$$\{v_1, \dots, v_j\} = A_0 \cup \{w_1, \dots, w_s\}.$$

Since $(\pi, j)$ is good for $(T_1, e)$, there is an embedding of $T_1$ into $H[A_0 \cup \{w_1, \dots, w_s\}]$ with root assignment $\psi$. By the definition of $\mathcal{A}$, $\{w_1, \dots, w_s\} \in \mathcal{A}$. Thus $(w, s)$ is counted by $c_U(\mathcal{A})$.

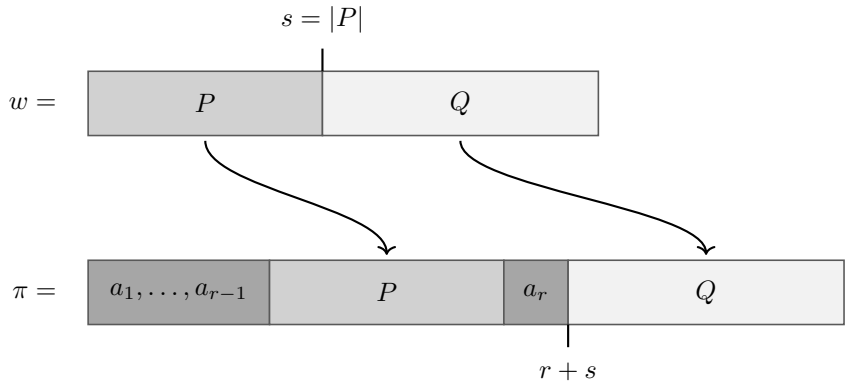



Figure 4: The bijection in (4.10).

Conversely, take a pair $(w, s)$ counted by $c_U(\mathcal{A})$. Form $\pi$ by placing $a_1, \dots, a_{r-1}$ first, followed by the first $s$ entries of $w$, then $a_r$, and then the remaining entries of $w$. Mark the position $j = r + s$, which is occupied by $a_r$. The resulting state belongs to $\Omega_{\mathbf{a}}$, and its first $j$ vertices form the set $A_0 \cup \{w_1, \dots, w_s\}$. Since $\{w_1, \dots, w_s\} \in \mathcal{A}$, the definition of $\mathcal{A}$ supplies an embedding of $T_1$ into the hypergraph induced by these vertices, with root assignment $\psi$. Hence the state is good for $(T_1, e)$.

These operations are inverse, proving $C_{\mathbf{a}}(T_1, e) = c_U(\mathcal{A})$. Replacing $T_1, \mathcal{A}$ by $T_2, \mathcal{B}$ or by $T, \mathcal{C}$ proves the other two equalities.

The same deletion and insertion operations apply to all states in $\Omega_{\mathbf{a}}$, whether or not they are good. They give the bijection

$$(w, s) \longleftrightarrow \big((a_1, \dots, a_{r-1}, w_{\le s}, a_r, w_{>s}),\ r + s\big), \qquad w \in \mathrm{Perm}(U), \quad 0 \le s \le m. \tag{4.10}$$

Figure 4 illustrates this correspondence. The case $s = 0$ is allowed and gives the marked position $j = r$. There are $m!$ choices for $w$ and $m + 1$ choices for $s$, so

$$|\Omega_{\mathbf{a}}| = m!(m+1) = (m+1)!.$$

We can now apply Lemma 4.3. Using the three identities proved above, we obtain

$$C_{\mathbf{a}}(T_1, e) + C_{\mathbf{a}}(T_2, e) = c_U(\mathcal{A}) + c_U(\mathcal{B}) \le (m+1)! + c_U(\mathcal{C}) = |\Omega_{\mathbf{a}}| + C_{\mathbf{a}}(T, e),$$

which is precisely (4.9). Finally, sum over all $\mathbf{a} \in \mathcal{I}_H$. By (4.8),

$$C(T_1, e) + C(T_2, e) \le \sum_{\mathbf{a} \in \mathcal{I}_H} \big(|\Omega_{\mathbf{a}}| + C_{\mathbf{a}}(T, e)\big) = r!|E(H)|(m+1)! + C(T, e) = M + C(T, e),$$

where the last equality follows from $m + 1 = n - r + 1$ and (3.3). □

### 4.4 Completing the proof of Theorem 2.1

In this section we obtain an upper bound for $M$ by applying Lemmas 4.2 and 4.4, and use this to complete the proof of Theorem 2.1.

**Theorem 4.5.** *For every tight $r$-tree $T$ with $t \geq 1$ edges and every $e \in E(T)$,*

$$M \leq C(T, e) + (t-1)W. \tag{4.11}$$

*Proof.* We proceed by induction on $t$. For $t = 1$, the claim is (4.2). Suppose $t \geq 2$, and apply Lemma 4.1 at $e$. In case (i), the smaller tight tree $S$ has $t-1$ edges and for some edge $f \in E(S)$ the edge $e$ intersects $f$ in exactly $r-1$ vertices. The induction hypothesis rooted at $f$, followed by Lemma 4.2, gives

$$M \leq C(S, f) + (t-2)W \leq C(T, e) + (t-1)W.$$

In case (ii), write $t_i = |E(T_i)|$. Then $t_i < t$ for $i = 1, 2$ and $t_1 + t_2 = t + 1$. Add the two induction inequalities and apply Lemma 4.4 to obtain

$$\begin{aligned} 2M &\leq C(T_1, e) + C(T_2, e) + (t_1 + t_2 - 2)W \\ &\leq M + C(T, e) + (t-1)W. \end{aligned}$$

Subtracting $M$ completes the induction. □

*Proof of Theorem 2.1.* If $E(H) = \varnothing$, the assertion is immediate. Otherwise $n \geq r$. Fix any edge $e$ of $T$. Since $H$ is $T$-free, $C(T, e) = 0$. Theorem 4.5 and (3.3) give

$$r!\,|E(H)|\,(n-r+1)! = M \leq (t-1)W = (t-1)(r-1)!\,|\partial H|\,(n-r+1)!.$$

Cancelling $(r-1)!(n-r+1)!$ proves $r|E(H)| \leq (t-1)|\partial H|$. Finally, $|\partial H| \leq \binom{n}{r-1}$ gives (2.2). □

## References


[1] M. Ajtai, J. Komlós, M. Simonovits, and E. Szemerédi, *Solution of the Erdős–Sós Conjecture*, presentation at the Bondy Conference, Montreal, 2003.

[2] G. Besomi, M. Pavez-Signé, and M. Stein, *Degree conditions for embedding trees*, SIAM J. Discrete Math. **33** (2019), 1521–1555.

[3] G. Besomi, M. Pavez-Signé, and M. Stein, *On the Erdős–Sós conjecture for trees with bounded degree*, Combin. Probab. Comput. **30** (2021), 741–761.

[4] S. Brandt and E. Dobson, *The Erdős–Sós conjecture for graphs of girth 5*, Discrete Math. **150** (1996), 411–414.

[5] A. Davoodi, D. Piguet, H. Řada, and N. Sanhueza-Matamala, *The asymptotic version of the Erdős–Sós conjecture and beyond*, arXiv:2603.17755, 2026.

[6] T. Adamczewski and T. F. Bloom, *FrontierMath Erdős*, Epoch AI report, September 2026. `https://epoch.ai/files/frontiermath-erdos.pdf`.

[7] P. Erdős, *Extremal problems in graph theory*, in: Theory of Graphs and its Applications (M. Fiedler, ed.), Academic Press, New York, 1965, 29–36.

[8] P. Erdős and T. Gallai, *On maximal paths and circuits of graphs*, Acta Math. Acad. Sci. Hungar. **10** (1959), 337–356.

[9] P. Erdős and V. T. Sós, *Some remarks on Ramsey's and Turán's theorem*, in: Combinatorial Theory and its Applications, II (Proc. Colloq., Balatonfüred, 1969), Colloq. Math. Soc. János Bolyai, Vol. 4, North-Holland, Amsterdam, 1970, pp. 395–404.

[10] G. Fan and L. Sun, *The Erdős–Sós conjecture for spiders*, Discrete Math. **307** (2007), 3055–3062.

[11] P. Frankl and Z. Füredi, *Exact solution of some Turán-type problems*, J. Combin. Theory Ser. A **45** (1987), no. 2, 226–262.

[12] Z. Füredi, T. Jiang, A. Kostochka, D. Mubayi and J. Verstraëte, *Hypergraphs not containing a tight tree with a bounded trunk*, SIAM J. Discrete Math. **33** (2019), no. 2, 862–873. arXiv:1712.04081.

[13] Z. Füredi, T. Jiang, A. Kostochka, D. Mubayi and J. Verstraëte, *Tight paths in convex geometric hypergraphs*, Adv. Combin. **2020** (2020), Paper No. 1, 14 pp. doi:10.19086/aic.12044.

[14] S. Glock, D. Kühn, A. Lo, D. Osthus, *The existence of designs via iterative absorption: hypergraph $F$-designs for arbitrary $F$*, Memoirs of the American Mathematical Society **284** (2023), monograph 1406.

[15] A. Goerlich and A. Żak, *On Erdős–Sós conjecture for trees of large size*, Electron. J. Combin. **23** (2016), Paper P1.52.

[16] F. Havet, B. Reed, M. Stein, and D. R. Wood, *A variant of the Erdős–Sós conjecture*, J. Graph Theory **94** (2020), 131–158.

[17] P. Keevash, *The existence of designs*, arXiv:1401.3665, 2014; revised 2024.

[18] Y. S. Kupitz, M. Perles, *Extremal theory for convex matchings in convex geometric graphs*, Discrete Comput. Geom. **15**, (1996), 195–220.

[19] A. McLennan, *The Erdős–Sós conjecture for trees of diameter four*, J. Graph Theory **49** (2005), 291–301.

[20] D. Mubayi and J. Verstraëte, *Counting trees in graphs*, Electron. J. Combin. **23** (2016), no. 3, Paper P3.39.

[21] A. Pokrovskiy, *Hyperstability in the Erdős–Sós conjecture*, arXiv:2409.15191, 2024.

[22] B. Reed and M. Stein, *The Erdős–Sós conjecture in dense graphs*, arXiv:2609.05417, 2026.

[23] V. Rödl, *On a packing and covering problem*, European J. Combin. **6** (1985), no. 1, 69–78.

[24] V. Rozhoň, *A local approach to the Erdős–Sós conjecture*, SIAM J. Discrete Math. **33** (2019), 643–664.

[25] J.-F. Saclé and M. Woźniak, *The Erdős–Sós conjecture for graphs without $C_4$*, J. Combin. Theory Ser. B **70** (1997), 367–372.

[26] M. Stein, *Tree containment and degree conditions*, in: A. M. Raigorodskii and M. T. Rassias (eds.), Discrete Mathematics and Applications, Springer Optimization and Its Applications, Springer, 2020, pp. 459–486.

[27] M. Stein, *Kalai's conjecture in $r$-partite $r$-graphs*, European J. Combin. **117** (2024), Paper No. 103827.

[28] G. Tiner and Z. Tomlin, *On the Erdős–Sós conjecture for $k = 9$*, Alabama J. Math. **45** (2022), 37–45.

[29] C. Wilson, *A Tight Lower bound on Trees in Graphs*, https://arxiv.org/abs/2512.14890.

[30] M. Woźniak, *On the Erdős–Sós conjecture*, J. Graph Theory **21** (1996), 229–234.